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# Task 1: Geometrically Weighted Terms in ERGMs

# Task 2: Estimation and interpretation of an ERGM

```{r}
Install packages from CRAN
#install.packages(c("network", "ergm", "manynet", "patchwork", "latticeExtra", "texreg",
"broom"))
Install autograph from GitHub develop branch
install.packages("remotes") # if not already installed
#remotes::install_github("stocnet/autograph", ref = "develop")
Loading libraries - stocnet suite
library(manynet)
library(autograph)
Loading libraries - sna suite
library(network)
library(ergm)
Loading libraries - plotting and presentation
library(ggraph)
library(patchwork)
library(latticeExtra) # improve the aesthetics of the MCMC diagnostics plots
library(texreg)
library(broom)
library(sna)
library(xtable)
library(ggplot2)
```

## 2.1

```{r}
Reading the adjacency matrices using manynet::read_matrix()
friendship <- read_matrix("Glasgow/f1.csv", header = FALSE)
demographic <- read.csv("Glasgow/demographic.csv", header = TRUE)

net <- network(as.matrix(friendship), directed = TRUE)
net %v% "gender" <- as.character(demographic[, "gender"])
set.seed(161)
this from the last assignment: permutations <- 5000
#estimate ERGM with parameters of tie existence and if that tie is same gender
model <- ergm(net ~ edges + nodematch("gender"), control = control.ergm(seed = 161))

summary(model)

#get conditional probs -----
#probability that a tie exists between two nodes
theta_edges <- coef(model)["edges"]
#additional chance a tie exists if the nodes are same gender
theta_match <- coef(model)["nodematch.gender"]

Log-odds of a tie between two same-gender students
logodds_same <- theta_edges + theta_match

Converted to probability (exp / (1 + exp))
prob_same <- plogis(logodds_same)

```

```
same for different gender tie (nodematch = 0)
logodds_diff <- theta_edges
prob_diff <- plogis(logodds_diff)

cat(sprintf(" P(tie | same gender) : %.4f (%.1f %%)\n", prob_same, prob_same * 100))
cat(sprintf(" P(tie | diff gender) : %.4f (%.1f %%)\n", prob_diff, prob_diff * 100))
cat(sprintf(" Ratio Same-Diff gender : %.4f \n", prob_same/prob_diff))

...

```

Both results hold strong evidence that edges are both sparse ( $\theta_{\text{edges}} = -5.0133$ ) and homophillic ( $\theta_{\text{match}} = 2.0098$ ), as  $p \ll 0.05$  in both cases. Although there is only a probability of  $P_{\text{same}} = 4.7\%$  for a tie to exist between two students of the same gender, it is still  $7.1563$  times higher than the probability of a tie existing between two students of opposite gender  $P_{\text{diff}} = 0.7\%$ . The huge difference in Null Deviance to Residual Deviance ( $22890 \rightarrow 3824$ ) also tells that the tie structure is very dependent on the chosen parameters.

## ## 2.2

To build the big ERGM model similar to 2.1:

```
```{r}
alcohol <- read.csv("Glasgow/alcohol.csv", header = TRUE)

# Attach just the first column
net %v% "alcohol" <- alcohol[, 1]

#this takes a lil while:
big_model <- ergm(net ~ edges + nodematch("gender") #same baseline as 2.1
  + mutual #(i) Reciprocity
  + gwesp(0.1, fixed = TRUE) #(ii) Transitivity

  + gwodegree(0.1, fixed = TRUE) # controls for out-degree distribution
  + gwidegree(0.1, fixed = TRUE) # controls for in-degree distribution

  + nodeofactor("gender", base = 2) #(iii) Boys have higher node outdegree
  + nodeifactor("gender", base = 2) #(iv) Boys have higher node indegree
  + nodeocov("alcohol") #(v) high alcohol -> high node outdegree
  + nodeicov("alcohol") #(vi) high alcohol -> high node indegree
  + absdiff("alcohol"), #(vii) alcohol homophily
  control = control.ergm(seed = 161, MCMLL.maxit = 40, MCMC.samplesize = 2048,
    MCMC.burnin = 10000, MCMC.interval = 500)
)
#summary(big_model)
#mcmc.diagnostics(big_model)

bigger_model <- ergm(net ~ edges + nodematch("gender") #same baseline as 2.1
  + mutual #(i) Reciprocity
  + gwesp(0.1, fixed = TRUE) #(ii) Transitivity

  + gwodegree(0.1, fixed = TRUE) # controls for out-degree distribution
  + gwidegree(0.1, fixed = TRUE) # controls for in-degree distribution

  + nodeofactor("gender", base = 2) #(iii) Boys have higher node outdegree
  + nodeifactor("gender", base = 2) #(iv) Boys have higher node indegree
  + nodeocov("alcohol") #(v) high alcohol -> high node outdegree
  + nodeicov("alcohol") #(vi) high alcohol -> high node indegree
  + absdiff("alcohol"), #(vii) alcohol homophily
  control = control.ergm(seed = 161, MCMC.samplesize = 50000, init =
coef(big_model),
  MCMC.burnin = 100000, MCMC.interval = 2000, MCMLL.maxit = 60)
)

summary(bigger_model)
mcmc.diagnostics(bigger_model)

...

```

2.3

The ERGM model here itself has the 7 parameters to check for the wanted features (α - γ). To achieve convergence we had to add in 2 more control parameters: $\gamma_{wdegree}$ and $\gamma_{gwidegree}$. We see in the summary, that both $\gamma_{wdegree}$ and $\gamma_{gwidegree}$ are significant, so they were important to explain the network and help with the convergence of the algorithm.

Using a high decay value like $\alpha = 0.3$ or $\alpha = 0.6$ gave problems while converging so we went for one on the lower bound of $\alpha = 0.1$, since the MCMC sampler tended to generate networks with way more edges than observed.

The low α decay value also tells that in the network, transitivity is only important over short distances, and longer chains do not contribute much.

Using small parameters in the control parameter, the `mcmc.diagnostics()` gave some bad results: The Mean was far away from 0 for many terms, the P-value was below or very close to 0.05 and the omnibus value being at 3.4×10^{-6} indicates the model has not fully converged yet.

All of this indicated that a longer MCMC run with larger parameters was needed as it is now the case.

So from this first model we now expand on it further by changing the parameters in control to larger samplesize, burnin and interval.

Further all this is done on the initial model as it just has not fully converged yet, and starting from these large settings gave a failure of convergence, as these are bad starting parameters.

This second ERGM fit may have only one iteration until it finishes, but it improves most statistics shown in `mcmc.diagnostics()`.

All terms now have a P-value above 0.05, but most are still not close to 1.

All individual Geweke statistics are within acceptable bounds and the Geweke joint P-value improved from 0.089 to 0.212, confirming chain stationarity.

The omnibus P-value is significantly bigger at $0.008 > 3.4 \times 10^{-6}$ but still below 0.05, which is fine since it is sensitive to large sample sizes and not indicative of systematic misfit in any specific statistic.

2.4

```
```{r}
gof_model <- gof(bigger_model, GOF = ~ idegree + odegree + espartners + distance) #+ dspartners

Plot auxiliary statistics
gof_model
```

```{r}
ergm:::plot.gof(gof_model)
```
```

Both the plot of `model$statistic` and `idegree` indicate a very good fit on these terms, as there are no outliers and their values are very close to the centre of the boxplots.

This confirms that the model is internally consistent and that it reproduces the in-degree distribution well, it accurately captures how many nominations students receive.

The `odegree` plot is somewhat predictably wrong, it overestimates the count of nodes that have exactly 1 outgoing edge and then has too few with those that have 3-6 edges.

It wrongly assumes that there are some nodes with more than 6 outgoing edges, which is not possible per the data collection, as that only allowed up to 6 friendship nominations per person, which is information that the model does not know about.

The `edges-wise shared partners` plot shows that the model has problems with predicting triads, which suggests that the chosen decay value $\alpha = 0.1$ might not have been the best choice.

The bad fitting of the `minimum goedesic distance` plot shows that the estimated model is more fragmented and less connected than the real observed data would suggest.

Overall, the model shows clear parts that can still be improved, for one there is the α decay value that could be tweaked until there are better results.

There could also be more parameters that the model does not count for yet, like the fact that a node has a maximum out-degree of 6.

It seems to be better with replicating local features, while its weaknesses lie with the global features.

2.5

```

```{r}
summary(bigger_model)
```

```

Terms not used for any hypothesis: The \$edges\$ coefficient of $(-5.95, p < 0.001)$ confirms that the friendship network is very sparse and the probability for a tie existing between any two students is very low.

Very logical for a huge network, where any student can only nominate a maximum of 6 friends. Both \$gwodegree\$ $(+1.84, p < 0.001)$ and \$gwidegree\$ $(+1.60, p < 0.001)$ are significant, confirming there is a clear difference between how many nominations students both send and receive.

Some students are far more social or popular than others.

The gender homophily from 2.1 still stays relevant here with \$nodematch.gender\$ $(+0.85, p < 0.001)$ and indicates that it plays an important role in how ties are formed.

It is to be noted though, that this term dropped from the model in 2.1 with a value of $+2.01$, so some of the networks model can be better explained by other terms introduced in this new bigger model.

Hypothesis (\$i\$) Reciprocity: \$mutual\$ $(+3.02, p < 0.001)$.

The hypothesis is strongly supported by these values.

If student j has nominated student i , the odds of i nominating j back are multiplied by $e^{3.02} \approx 20.5$.

This is the strongest effect that is found in this model, indicating it is a crucial deciding factor in how nominations are built up on.

Hypothesis (\$ii\$) Transitivity: \$gwes\$ $(+1.68, p < 0.001)$.

The hypothesis is strongly supported by these values.

When two students share a common friend, a tie between them becomes significantly more likely. This supports a typical triadic closure structure in the model, which occurs in most social networks and also leads to the formation of clusters.

Hypothesis (\$iii\$) Boys have higher social activity: \$nodeofactor.gender\$ $(-0.10, p = 0.498)$. This hypothesis is not supported by these values.

The coefficient for boys sending more nominations than girls is small and negative, which would suggest that the opposite is slightly true.

This however is not the case either, since the P-value is so high and as such, not making it statistically significant.

Hypothesis (\$iv\$) Boys have higher popularity: \$nodeifactor.gender\$ $(+0.04, p = 0.800)$.

This hypothesis is not supported by these values.

The coefficient is near zero and the high P-value means it is completely insignificant.

Boys do not receive significantly more nominations than girls in this network, which, along with Hypothesis (\$iii\$), means that gender has no direct influence on how popular or socially active a person is.

Hypothesis (\$v\$) Alcohol consumption increases social activity: \$nodeocov.alcohol\$ $(-0.12, p = 0.032)$.

This hypothesis is not supported by these values, the somewhat low P-value along a negative coefficient say the opposite is supported.

Higher alcohol consumption is associated with fewer outgoing nominations.

A one unit increase in alcohol score decreases the odds-ratio of sending a tie by $e^{0.12} \approx 1.127$, suggesting that heavier drinkers are actually less likely to actively nominate friends, contrary to the hypothesis.

With four units difference from 1 to 5, this means an approximate odd-ratio decrease of $e^{0.12 * 4} \approx 1.616$ for the heaviest drinkers in comparison to those with a value of 1.

Hypothesis (\$vi\$) Alcohol consumption increases popularity: \$nodeicov.alcohol\$ $(+0.16, p = 0.004)$.

The hypothesis is supported by these values.

Students with higher alcohol consumption receive significantly more friendship nominations.

A one unit increase in alcohol score multiplies the odds of receiving a tie by $e^{0.16} \approx 1.174$, suggesting heavier drinkers are perceived as more popular or socially attractive by their peers.

With four units difference from 1 to 5, this means an approximate odd increase of $e^{0.16 * 4} \approx 1.896$ for the heaviest drinkers in comparison to those with a value of 1.

Hypothesis (\$vii\$) Alcohol homophily: \$absdiff.alcohol\$ $(-0.11, p = 0.003)$.

The hypothesis is supported by these values.

The negative coefficient on absolute difference means ties are more likely between students

with similar alcohol consumption levels.

The smaller the difference in alcohol scores between two students, the more likely they are to form a friendship, indicating clear homophilic tendencies regarding alcohol consumption.

2.6

Since the data collection method only allowed the students to nominate 6 friends, a constraint that enforces this could be added to the ERGM estimation model to get this point correct, which would make the model itself more related to the data and fit some terms better.

This would however raise the question if this is really wanted, since it is very possible that in real-life, students have more than 6 friendship nominations.

So this should rather be a problem to think about at the data collection part of data modelling.

From the results we found in 2.4, we should tweak the α decay value to a higher value like 0.3 or maybe even higher than that, since transitivity between students' friendships seems to hold over more nodes than the current value estimates for and it does not correctly simulate the triadic patterns that can be found in this network.

There is a very arguable point to be made that ties are also dependent on age-homophily, something that the current model does not account for at all, since the edges do not even have the age value added to them.

There is really no reason not to add the age parameter to the model, since we already have the data available.

Adding these terms:

```
```{r}
+ nodecov("age") + nodeicov("age") + absdiff("age")
```
```

could help the model.

Statistical outliers, like people who have to repeat a class, may be on one end of the spectrum and more likely to have drastically different data values than other students.

2.7

We suggest two hypotheses that center around the uncollected data of the students' grades.

The first hypothesis is centered around Academic homophily.

This would require the following addition to the model:

```
```{r}
+ absdiff("grade")
```
```

This could be computed like this: $\text{absdiff}(\text{"grade"}) = \sum_{i \neq j} y_{ij} \cdot |x_i^{\text{grade}} - x_j^{\text{grade}}|$, where $y_{ij}=1$ if student i has nominated student j and is 0 otherwise.

x_k^{grade} corresponds to the grade of student k .

A negative coefficient would support the hypothesis in this case, as it would make ties more probable if they have similar grade levels.

In the network graph, this effect would be seen as more ties existing between students of similar academic level.

It should also be noted, that since we just assume one total grade for each student, there might be different results if the same hypothesis is executed on just single subject grades.

The second hypothesis is that students with higher grades receive more friendship nominations. That would require this addition to the model to be figured out if the hypothesis is supported or not:

```
```{r}
+ nodeicov("grade")
```
```

This could be computed like this: $\text{nodeicov}(\text{"grade"}) = \sum_{i \neq j} y_{ij} \cdot x_j^{\text{grade}}$.

A positive coefficient would support this hypothesis, as that would mean that students with higher grades receive more nominations.

In the network graph, this could be seen as nodes with a high grade value having a relatively higher indegree than those with a lower grade.

Both of these hypotheses could explain some residual variation in tie formation, that has not yet been captured by our current model.

Task 3: Estimation and interpretation of SAOMs

3.1

3.1.1

```
```{r}
install.packages("RSiena")
library(RSiena)
library(parallel)
library(sna)

Load friendship waves
f1 <- matrix(as.numeric(as.matrix(read_matrix("Glasgow/f1.csv", header = FALSE))),
 nrow = 129, ncol = 129)
f2 <- matrix(as.numeric(as.matrix(read_matrix("Glasgow/f2.csv", header = FALSE))),
 nrow = 129, ncol = 129)
f3 <- matrix(as.numeric(as.matrix(read_matrix("Glasgow/f3.csv", header = FALSE))),
 nrow = 129, ncol = 129)

Read demographic, alcohol and logdistance data
demographic <- read.csv("Glasgow/demographic.csv", header = TRUE)
alcohol_data <- read.csv("Glasgow/alcohol.csv", header = TRUE)
logdist <- matrix(as.numeric(unlist(strsplit(readLines("Glasgow/logdistance.csv"), ","))), nrow
= 129, ncol = 129)

friendship <- as_dependent_rsiena(array(c(f1, f2, f3), dim = c(129, 129, 3)))

Create covariates
gender <- as_covariate_rsiena(demographic$gender, type = "monadic")
alcohol <- as_covariate_rsiena(alcohol_data[, 1], type = "monadic")
logdistance <- as_covariate_rsiena(logdist, type = "oneMode")

Combine into siena data object
mydata <- make_data_rsiena(friendship, gender, alcohol, logdistance)
#print(mydata)

Compute Jaccard index and data description
write_report(mydata, outputName = "glasgowJaccardIndex")
```

...

From the data generated in the file "glasgowJaccardIndex", we have the resulting Jaccard Index values from Wave 1  $\rightarrow$  Wave 2 with a value of  $0.304$  and from Wave 2  $\rightarrow$  Wave 3 with a value of  $0.351$ .

Both of these values fit neatly into the recommended range of  $[0.2, 0.6]$ , indicating that the data contains sufficient data to estimate a SAOM. If the value were below that range, it would mean the network changes too drastically and leaves no network stability, while a value above the range would indicate that the network hardly changes at all and shows no evolutionary traits. The higher Jaccard index from Wave 2  $\rightarrow$  Wave 3 indicates that the network gained stability over time.

### 3.1.2

```
```{r}
myeff <- make_specification(mydata)

#endogenous structural effects, from what generally should hold for networks
myeff <- set_effect(myeff, list(recip))      # reciprocity
myeff <- set_effect(myeff, list(transTrip))  # transitivity
myeff <- set_effect(myeff, list(cycle3))     # 3-cycles

#exogenous control effects, we know from task 2 that gender plays a huge role
myeff <- set_effect(myeff, list(egoX, altX, sameX), covar1 = "gender")

#Hypothesis i: popularity
myeff <- set_effect(myeff, list(inPopSqrt))

#Hypothesis ii: alcohol homophily
```

```

myefff <- set_effect(myefff, list(egoX, altX, simX), covar1 = "alcohol")

#Hypothesis iii: geographic proximity
myefff <- set_effect(myefff, list(X), covar1 = "logdistance")

myefff
```
3.1.3

```{r}
SAOMAlgorithm <- set_algorithm_saom(seed = 161, n3 = 3000, nsub = 4)

# Estimate the model
model_saom <- siena(data = mydata,
                    effects = myefff,
                    returnDeps = TRUE,
                    batch = FALSE,
                    control_algo = SAOMAlgorithm)

#Check results and convergence criterion (as presented in the lecture slides)
#model_saom
t_ratios <- apply(model_saom$sf, 2, mean) / apply(model_saom$sf, 2, sd)
t_ratios

overall <- sqrt(t(apply(model_saom$sf, 2, mean)) %*% solve(cov(model_saom$sf)) %*%
               apply(model_saom$sf, 2, mean))
overall
```

```

As we can see from these results, we have that all absolute values of the t-ratios are below \$0.1\$, along with the overall value being below \$0.25\$. This fulfills the convergence criterion, indicating that all of our chosen parameters were sufficient to achieve convergence. The SAOM was estimated with 4 subphases and 3000 iterations in phase 3, which was enough to achieve this convergence and did not need to have another re-estimation or other model adjustments.

These parameters consist of the three needed for answering our hypotheses: One parameter to control for the popularity of a student by checking for its indegree value, another to check for alcohol consumption homophily and the last one to see if geographic proximity has an effect on the model evolution.

Furthermore we have some basic endogenous and exogenous variables that normally play a big part in the structure of most networks: Variables to check for the reciprocity of ties, the transitivity of ties and 3-cyclic patterns.

Lastly we added in one more parameter, to give control for gender homophily. We have seen in task 2 that it plays a huge role in the structure of this specific network.

```

3.1.4

```{r}
#most of this is just from the practical:
cl <- makeCluster(4)

# Indegree distribution
gof_id <- test_gof(
  model_saom,
  verbose = FALSE,
  varName = "friendship", IndegreeDistribution,
  cluster = cl
)

# Outdegree distribution
gof_od <- test_gof(
  model_saom,
  verbose = FALSE,
  varName = "friendship", OutdegreeDistribution,
  cluster = cl
)

# Triad census
gof_tc <- test_gof(

```

```

    model_saom,
    verbose = FALSE,
    varName = "friendship", TriadCensus,
    cluster = cl
)

# Geodesic distance
GeodesicDistribution <- function(
  i, data, sims, period, groupName,
  varName, levls = c(1:5, Inf), cumulative = TRUE
) {
  x <- networkExtraction(i, data, sims, period, groupName, varName)
  require(sna)
  a <- sna::geodist(symmetrize(x))$gdist
  if (cumulative) {
    gdi <- sapply(levls, function(i) { sum(a <= i) })
  } else {
    gdi <- sapply(levls, function(i) { sum(a == i) })
  }
  names(gdi) <- as.character(levls)
  gdi
}

gof_gd <- test_gof(
  model_saom,
  verbose = FALSE,
  varName = "friendship", GeodesicDistribution
)

stopCluster(cl)

#Plot all four
plot(gof_id)
plot(gof_od)
plot(gof_gd)
plot(gof_tc, center = TRUE, scale = TRUE)
gof_id
gof_od
gof_gd
gof_tc
\,\,\,

```

The plot for the goodness of fit for the indegree distribution shows that our SAOM model captures this property well. The plot follows the real data and as such produces a p-value of \$0.654\$, which supports that the model captures this property well, since high p-values support the property being captured.

The other three plotted parameters are all bad. To start of with, they all have a p-value of \$0\$, which means that not a single one of them accurately captures the observed data.

The outdegree distribution too few nodes with a low outdegree, and overcompensates for the ones with high outdegree, meaning that the model makes the people more popular than they are supposed to be and produces less people who are less popular.

The geodesic distribution itself has values that are way too low. All people are closer connected than they are in the observed data, meaning that

3.1.5

3.2

3.3

3.4

